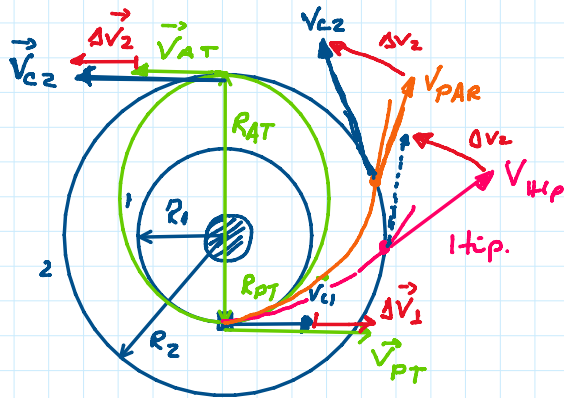


16/11/21.

Manobra Homann (1927)



0. Circular 1: $V_{c1} = \sqrt{\frac{GM}{R_1}}$

0. Circular 2: $V_{c2} = \sqrt{\frac{GM}{R_2}}$

2 corpos —

Circular	X	
Elíptica	✓	*
Parabólica	✓	
Hiperbólica	✓	

Eq. vis-viva $V = \sqrt{GM \left(\frac{2}{R} - \frac{1}{a} \right)}$

Raio do pericentro da elipse de transferência $R_{PT} = R_1$
 " " apocentro " " " " " " $R_{AT} = R_2$

Veloc. pericentro: da elipse de transferência

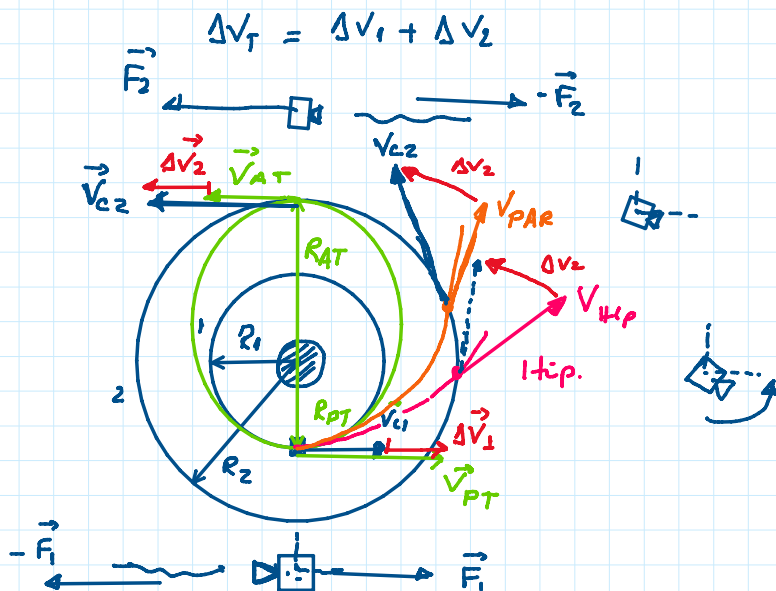
$$V_{PT} = \sqrt{GM \left(\frac{2}{R_1} - \frac{1}{a_T} \right)}$$

$$a_T = \frac{R_{PT} + R_{AT}}{2} = \frac{R_1 + R_2}{2}$$

$$V_{AT} = \sqrt{GM \left(\frac{2}{R_2} - \frac{1}{a_T} \right)}$$

Logo: $\Delta V_1 = V_{PT} - V_{c1} = \sqrt{GM \left(\frac{2}{R_1} - \frac{1}{a_T} \right)} - \sqrt{\frac{GM}{R_1}} = \dots$

$$\Delta V_2 = V_{c2} - V_{AT} = \sqrt{\frac{GM}{R_2}} - \sqrt{GM \left(\frac{2}{R_2} - \frac{1}{a_T} \right)} = \dots$$



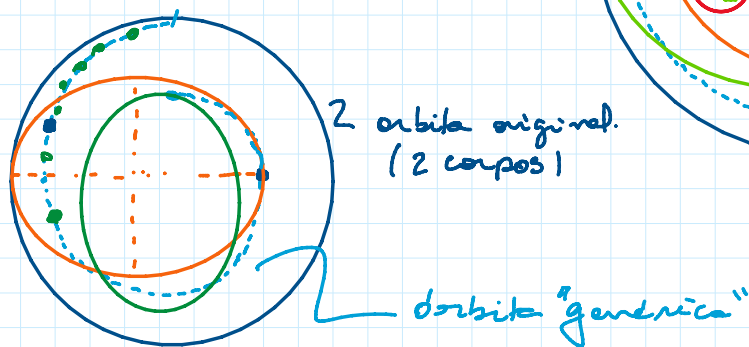
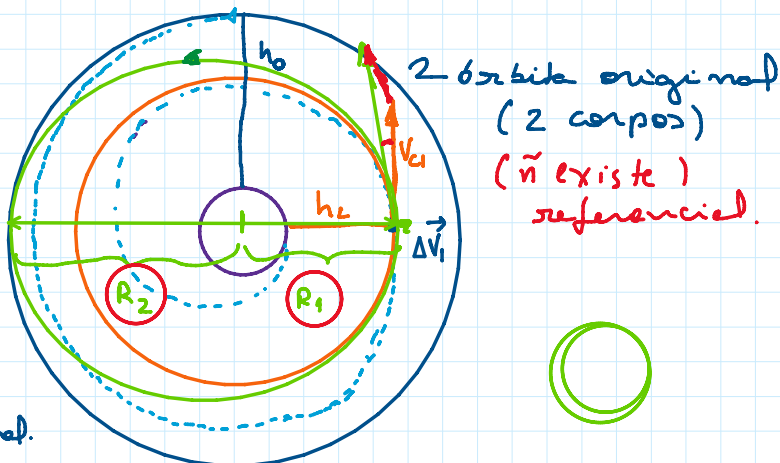
← →



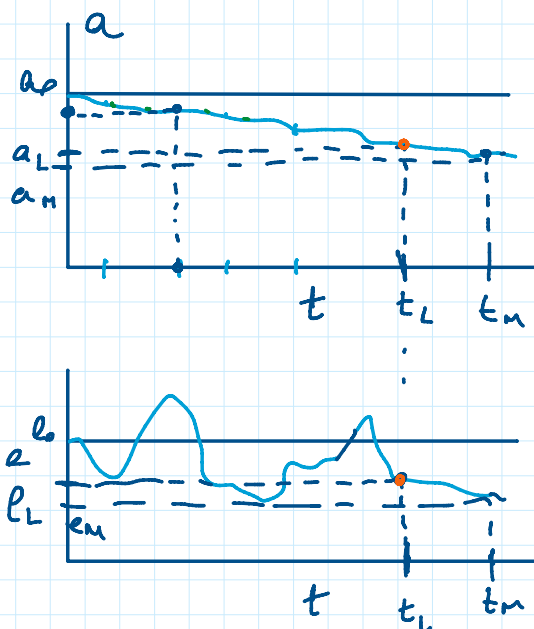
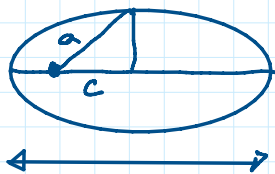
Onde vamos usar?

Situação.

Órbita osciladora?

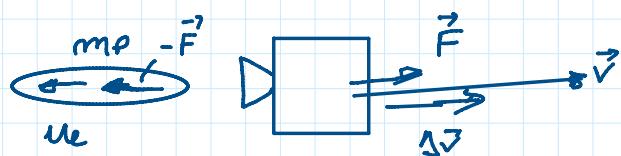


$$e = \frac{c}{a}$$



Equação de Tsiolkovsky

Permite calcular a massa de propelente gasta em uma manobra.



$$m_p = m_i \left[1 - \exp\left(-\frac{\Delta V}{I_{sp} g_0}\right) \right]$$

I_{sp} = impulso específico
s.

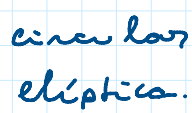
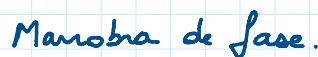
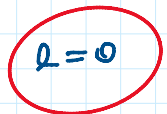
$$I_{sp} = \frac{U_e}{g_0}$$

1º impulso p/ gerar ΔV_1 . $m_{p1} = m_i \left[1 - \exp\left(-\frac{\Delta V_1}{I_{sp1} g_0}\right) \right]$

2º impulso p/ gerar ΔV_2 . $m_{p2} = (m_i - m_{p1}) \left[1 - \exp\left(-\frac{\Delta V_2}{I_{sp2} g_0}\right) \right]$

$$m_p = m_{p1} + m_{p2}.$$





$$-\Delta \hat{V}_I = V_C - V_A$$

$$V_A = \sqrt{GM \left(\frac{2}{R_e} - \frac{1}{a_I} \right)}$$

$$a_I = ?$$



$$\frac{-e_r^2}{r \cos \nu_c}$$